Geometry
Unit 12
Lesson 1 Homework

Name___Answer Key_______
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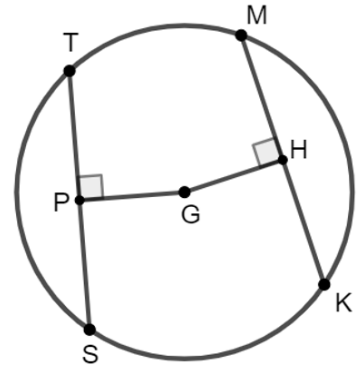
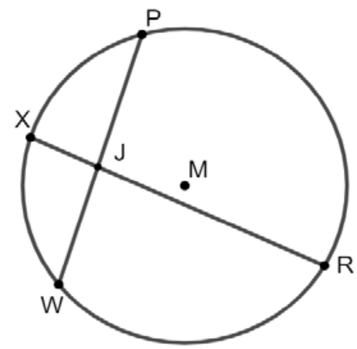
Circle G with points $T, M, K,$ and S on the circle are shown with chords $\overline{TS}$ and $\overline{MK}$ equidistant from the center. $TS = (12x - 5)$, $MK = (8x + 13)$, and $m\angle TPG = m\angle MHG = 90°$.



1. Find the value of x.
   $12x - 5 = 8x + 13$
   $4x = 18$
   $x = 4.5$

2. What is the length of chord $\overline{TS}$?
   $12(4.5) - 5 = TS = 49$

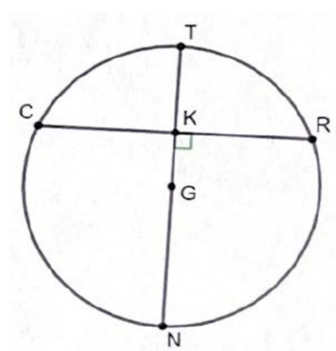
In Circle M, chords $\overline{RX}$ and $\overline{PW}$ intersect at point J. Points $R, W, X,$ and P lie on Circle M. $RJ = 4y$, $JX = 3$, $PJ = (2y - 2)$, and $JW = 8$.



3. Solve for the value of .
   $3(4y) = 8(2y - 2)$
   $12y = 16y - 16$
   $4y = 16$
   $y = 4$
4. What is the length of $\overline{PJ}$?
   $2(4) - 2 = PJ = 6$

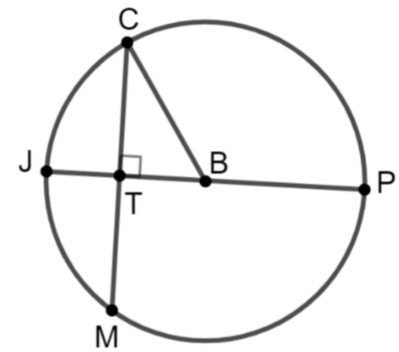
5. Find the length of chord $\overline{PW}$.
   $PJ + JW = 6 + 8 = PW = 14$

Circle G is shown where points $C, T, R,$ and N are on the circle. The diameter, $\overline{NT}$, is a perpendicular bisector of chord $\overline{CR}$. $CR = (13x + 5)$ and $KR = (25x - 1)$.



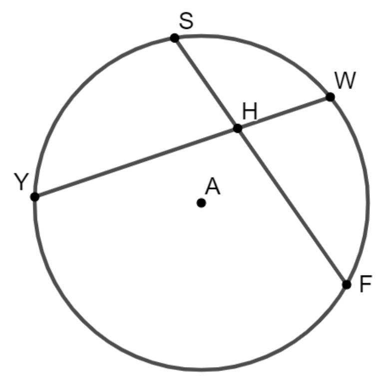
6. Calculate the value of x.
   $2(25x - 1) = 13x + 5$
   $50x - 2 = 13x + 5$
   $37x = 7$
   $x = \frac{7}{37} \approx 0.189$
7. What is the length of chord $\overline{CR}$?
   $13\left(\frac{7}{37}\right) + 5 = CR = \frac{276}{37} \approx 7.459$

Chord $\overline{CM}$ is drawn in Circle B so that it intersects the diameter $\overline{JP}$.



8. Given that $CM = 30$ and $TB = 8$, what is the length of $\overline{BP}$?
$BP = BC = \sqrt{(CT)^2 + (TB)^2} = \sqrt{(15)^2 + (8)^2} = 17$

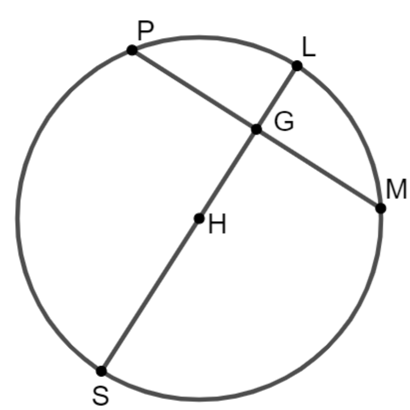
Circle A is shown where points $Y, S, W,$ and F lie on the circle. Chords $\overline{SF}$ and $\overline{YW}$ intersect at point H. $YH = 8$, $SH = 6$, $HW = (5x - 3)$, and $HF = (2x + 10)$. Aniyah and Colton are solving for x and their work is shown below.



Aniyah's Work	**Colton's Work**
$8(5x - 3) = 6(2x + 10)$ $40x - 24 = 12x + 60$ $28x = 84$ $x = 3$	$6(5x - 3) = 8(2x + 10)$ $30x - 18 = 16x + 80$ $14x = 98$ $x = 7$

9. Whose work do you agree with? Justify your answer.
Aniyah is correct. Colton did not multiply the correct segment lengths.

In Circle H, points $P, L, M,$ and S lie on the circle. Chord $\overline{PM}$ intersects the diameter $\overline{SL}$ at point G, and $m\angle HGP = 90°$.



10. If $PM = 24$ and $HS = 14$, what is the length of $\overline{GH}$? Round to the nearest hundredth if necessary.
$GH = \sqrt{(14)^2 - (12)^2} = \sqrt{52} \approx 7.21$

Geometry
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Lesson 2 Homework

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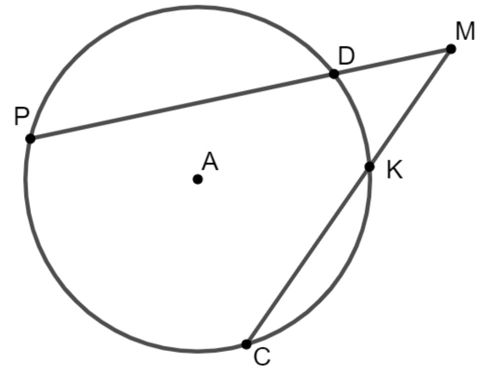
Circle A with secants $\overline{PM}$ and $\overline{CM}$ is shown. $\overline{PM}$ and $\overline{CM}$ intersect at point M outside the circle so that $PD = x$, $DM = 3$, $CK = 8$, and $KM = 4$.

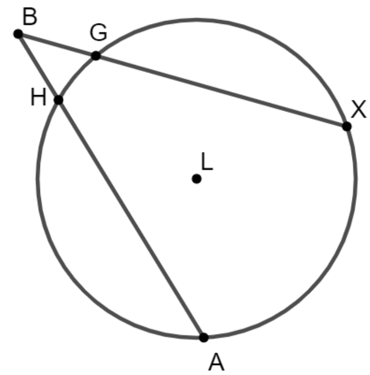


1. What is the length of secant $\overline{CM}$?
   $8 + 4 = CM = 12$

2. Find the value of x.
   $3(x + 3) = 4(8 + 4)$
   $3x + 9 = 48$
   $3x = 39$
   $x = 13$

3. What is the length of secant $\overline{PM}$?
   $(13) + 3 = PM = 16$

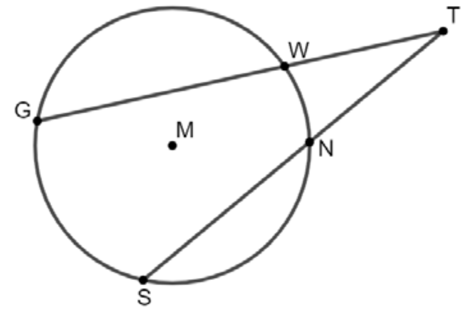
Circle L is shown, where points G, X, A, and H lie on circle L. Secants $\overline{BX}$ and $\overline{BA}$ intersect at point B outside the circle such that $BG = 6$, $GX = 20$, $BH = 4$. and $HA = (6x - 7)$.



4. What is the value of x?
   $6(20 + 6) = 4(6x - 7 + 4)$
   $156 = 24x - 12$
   $24x = 168$
   $x = 7$

5. Calculate the length of secant $\overline{BA}$.
   $4 + [6(7) - 7] = BA = 39$

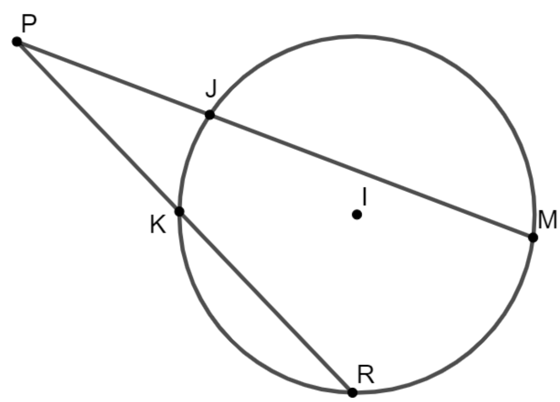
Circle M with secants $\overline{GT}$ and $\overline{ST}$ is shown. Secants $\overline{GT}$ and $\overline{ST}$ intersect outside the circle at point T so that $GW = 12x$, $WT = 4x$, $SN = 20$, and $NT = 16$.



6. Find the value of x.
   $4x(16x) = 16(36)$
   $64x^2 = 576$
   $x^2 = 9$
   $x = 3$

7. What is the length of secant $\overline{GT}$?
   $12(3) + 4(3) = GT = 48$

Carter and Isabella were solving for x in circle I where points J, M, R, and K are on the circle and secants $\overline{PM}$ and $\overline{PR}$ intersect at point P outside the circle. They know $PJ = 3$, $JM = (4x - 1)$, $PK = 4$, and $KR = 2x$.



Each of their equations are set up to solve for x.

Carter's Work	Isabella's Work
$3(4x - 1) = 4(2x)$	$3(4x + 2) = 4(2x + 4)$

8. Whose work is the correct equation needed to solve for x?
   Isabella's

9. Use this equation to find the value of x.
   $3(4x + 2) = 4(2x + 4)$
   $12x + 6 = 8x + 16$
   $4x = 10$
   $x = 2.5$

10. What is the length of secant $\overline{PR}$?
    $4 + 2(2.5) = PR = 9$

Geometry
Unit 12
Lesson 3 Homework

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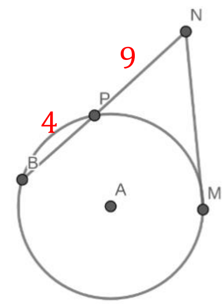
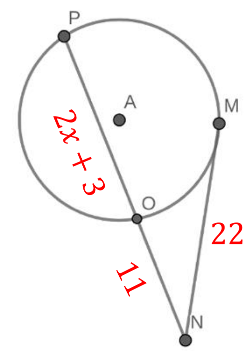
1. Circle A is shown with points M, B and P on the circle. Secant $\overline{BN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PB = 4$ and $PN = 9$. What is the length of $\overline{MN}$? Round to the nearest thousandth.



$9(13) = (MN)^2$

$\boxed{MN \approx 10.817}$

2. Circle A is shown with points M, O and P on the circle. Secant $\overline{PN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PO = (2x + 3)$, $ON = 11$ and $MN = 22$. Find the value of x.

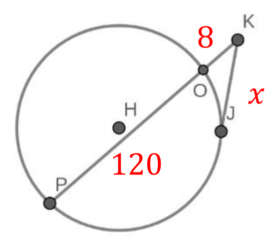


$11(2x + 3 + 11) = 22^2$
$22x + 154 = 484$
$22x = 330$
$\boxed{x = 15}$

3. What is the length of $\overline{PN}$?

$2(15) + 3 + 11 = \boxed{PN = 44}$

4. Circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 120$, $OK = 8$ and $KJ = x$. Find the value of x.



$8(128) = x^2$

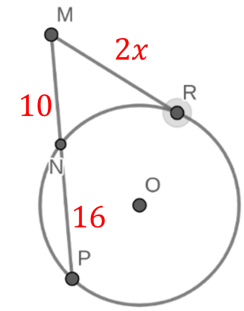
$\boxed{x = 32}$

5. Using the same image as question 4, circle H is shown with points J, O, and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 12 + x$, $OK = 10$ and $KJ = 18$. Find the length of $\overline{KP}$.

$10\big((12 + x) + 10\big) = 18^2$
$10x + 220 = 324$
$10x = 104$
$x = 10.4$

$\big(12 + (10.4)\big) + 10 = \boxed{KP = 32.4}$

6. Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = 16$, $MN = 10$ and $MR = 2x$. What is the value of x? Round to the nearest thousandth.

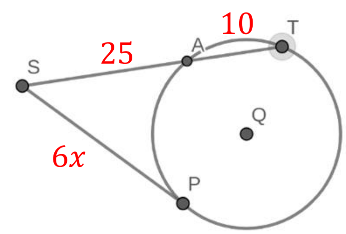


$10(26) = (2x)^2$
$4x^2 = 260$
$x^2 = 65$
$\boxed{x \approx 8.062}$

7. Using the same image as question 6, Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = x + 17$, $MN = 12$ and $MR = 24$. What is the value of x?

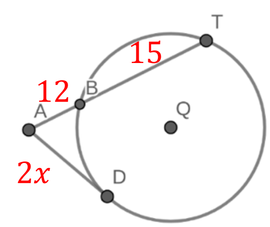
$12(x + 17 + 12) = 24^2$
$12x + 348 = 576$
$12x = 228$
$\boxed{x = 19}$

8. Circle Q is shown with points A, T and P on the circle. Secant $\overline{ST}$ and tangent $\overline{SP}$ intersect at point S outside the circle. $PS = 6x$, $AT = 10$ and $AS = 25$. What is the value of x? Round to the nearest thousandth.



$25(35) = (6x)^2$
$36x^2 = 875$
$x^2 \approx 24.306$
$\boxed{x \approx 4.930}$

9. Daniella and Taylor are working on finding the length of $\overline{AD}$. Circle Q is shown with points B, T and D on the circle. Secant $\overline{AT}$ and tangent $\overline{AD}$ intersect at point A outside the circle. $BT = 15$, $AB = 12$ and $AD = 2x$. Their partial work solving for x is shown below. Whose work do you agree with? Taylor's



Daniella's work	Taylor's work
$(2x)^2 = 15(15+12)$ $4x^2 = 15(27)$ $4x^2 = 405$ $x^2 = 101.25$	$(2x)^2 = 12(15+12)$ $4x^2 = 12(27)$ $4x^2 = 324$ $x^2 = 81$

10. Using the image in Question 9, what is the length of $\overline{AD}$?

$2(9) = AD = 18$

$x = \sqrt{81}$
$x = 8$

Geometry
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Lesson 4 Homework

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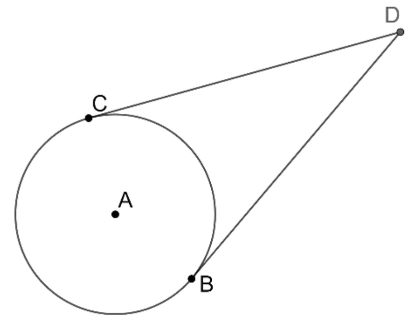
For questions 1 through 4, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
$CD = 10x + 30$ and $DB = 4x + 66$.



1. What is the value of x?

$$10x + 30 = 4x + 66$$
$$6x = 36$$
$$x = 6$$

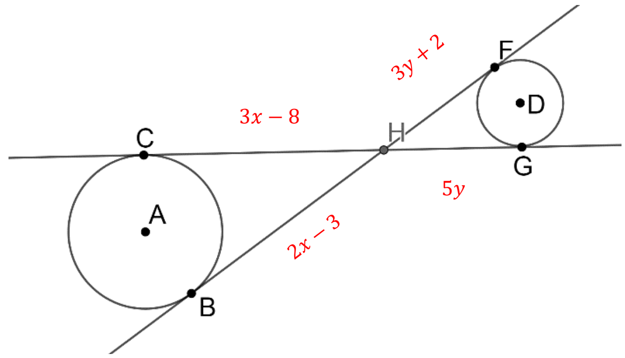
2. What is the length of $\overline{CD}$?

$$CD = 10(6) + 30 = 90$$

3. What is the length of $\overline{DB}$?

$$DB = 4(6) + 66 = 90$$

For questions 4 through 10, Circles A and D are shown. Lines $\overline{CG}$ and $\overline{BF}$ intersect at point H. Line $\overline{CG}$ is tangent to circle A at point C and circle D at point G. Line $\overline{BF}$ is tangent to circle A at point B and circle D at point F. $CH = 3x - 8, HG = 5y,$ $BH = 2x - 3, HF = 3y + 2$.



4. Write an equation you can use to solve for x.

$$3x - 8 = 2x - 3$$

5. Solve the equation you wrote in question 4 for x.

$$3x - 8 = 2x - 3$$
$$x = 5$$

6. Write an equation you can use to solve for y.

$$3y + 2 = 5y$$

7. Solve the equation you wrote in question 6 for y.

$$3y + 2 = 5y$$
$$2 = 2y$$
$$y = 1$$

8. What is the length of $\overline{BF}$?

$$BF = 2(5) - 3 + 3(1) + 2 = 12$$

9. What is the length of $\overline{CH}$?

$$CH = 3(5) - 8 = 7$$

10. What is the length of $\overline{HF}$?

$$HF = 3(1) + 2 = 5$$

Geometry
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Lesson 5 Homework

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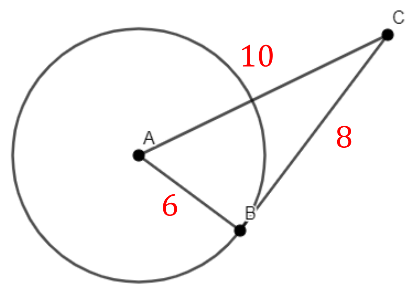
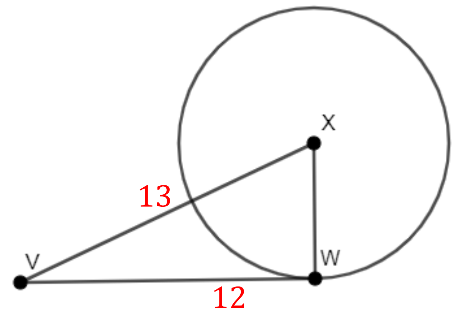
1. Circle A is shown with point C outside the circle, where point B is on the circle, $AC = 10$, $CB = 8$, and $BA = 6$. Verify if $\overline{BC}$ is tangent to circle A using the Pythagorean Theorem.



$6^2 + 8^2 = 10^2$
$36 + 64 = 100$
$100 + 100$ ✓ $\Longrightarrow$ TANGENT

Circle X on the right has $\overline{VW}$ as a tangent segment where W is the point of tangency. $VX = 13$ and $VW = 12$.



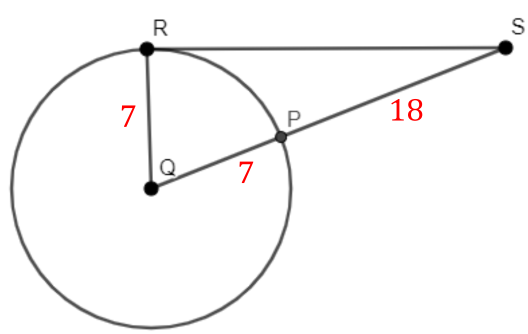
2. What is the length of $\overline{XW}$?

$12^2 + (XW)^2 = 13^2$
$(XW)^2 = 25$
$\boxed{XW = 5}$

3. What is the area of ΔXWV?

$\frac{1}{2}(12)(5) = \boxed{30}$

In Circle Q, $\overline{RS}$ is tangent to the circle at point R, $QR = 7$, and $PS = 18$.



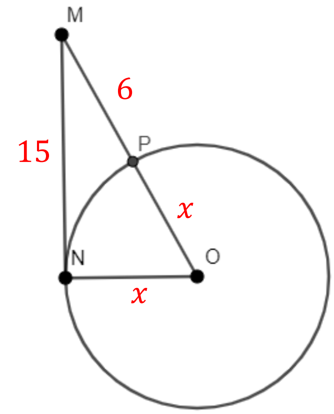
4. What is the length of $\overline{RS}$?

$(RS)^2 + 7^2 = (7 + 18)^2$
$(RS)^2 + 49 = 625$
$(RS)^2 = 576$
$\boxed{RS = 24}$

5. What is the area of ΔQRS?

$\frac{1}{2}(7)(24) = \boxed{84}$

In Circle O, $\overline{MN}$ is a tangent segment whose point of tangency is point N. $MN = 15$ and $MP = 6$.



6. What is the length of $\overline{ON}$?

$x^2 + 15^2 = (6 + x)^2$
$x^2 + 225 = 36 + 12x + x^2$
$12x = 189$
$x = \boxed{ON = 15.75}$

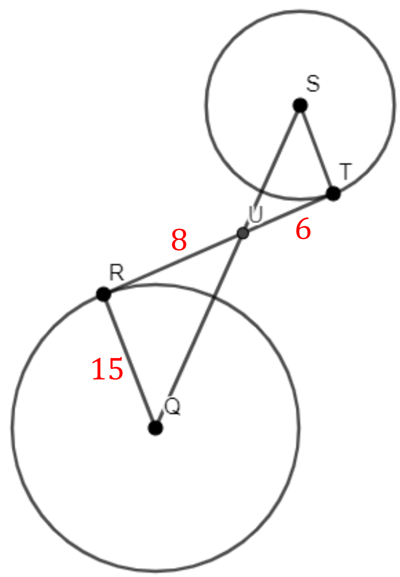
7. What is the length of $\overline{MO}$?

$6 + (15.75) = \boxed{MO = 21.75}$

8. What is the perimeter of ΔMNO?

$15 + 15.75 + 21.75 = \boxed{52.5}$

In the figure on the right, $\overline{RT}$ is a common tangent segment of both Circle Q and Circle S. $QR = 15$, $RU = 8$, and $TU = 6$.



9. What is the length of $\overline{ST}$?

$\frac{8}{6} = \frac{15}{ST}$

$8(ST) = (6)(15)$

$\boxed{ST = 11.25}$

10. What is the length of $\overline{QS}$?

$\overline{QU} = \sqrt{8^2 + 15^2} = 17$

$\overline{US} = \sqrt{6^2 + (11.25)^2} = 12.75$

$\overline{QS} = 17 + 12.75$

$\boxed{QS = 29.75}$

Geometry
Unit 12
Lesson 6 Homework

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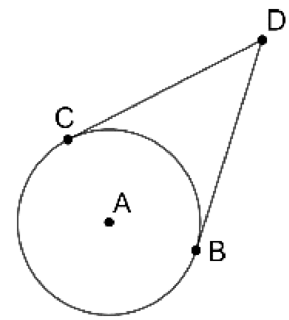
For questions 1 through 3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
$CD = 3x + 8$ and $DB = 5x - 16$.



1. What is the value of x?

$3x + 8 = 5x - 16$
$24 = 2x$
$12 = x$

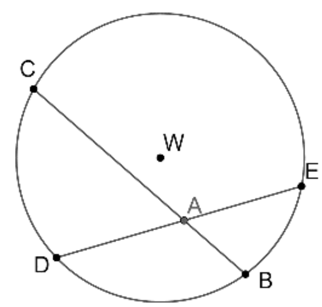
2. What is the length of $\overline{CD}$?

$3(12) + 8 = 44$

3. What is the length of $\overline{DB}$?

$5(12) - 16 = 44$

For questions 4 through 7, Circle W is shown. Chords $\overline{CB}$ and $\overline{DE}$ intersect at point A. $CA = 3x + 5$, $AB = 3$, $DA = 4$, and $AE = 2x + 4$.



4. Write an equation you can use to solve for x.

$3(3x + 5) = 4(2x + 4)$

5. Solve the equation you wrote in question 4 for x.

$9x + 15 = 8x + 16$
$x = 1$

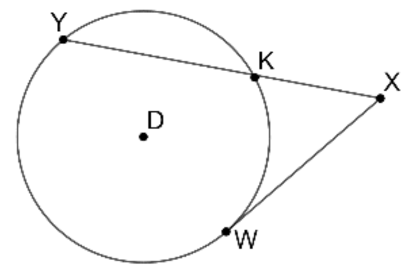
6. What is the length of $\overline{CB}$?

$3(1) + 5 + 3 = 11$

7. What is the length of $\overline{DE}$?

$2(1) + 4 + 4 = 10$

For questions 8 through 10, Circle D is shown. $\overline{XY}$ is a secant to the circle, and intersects tangent $\overline{XW}$ at point X. The length of $\overline{XW} = 6$, $\overline{KX} = 2$, and $\overline{YX} = 3x + 1$.



8. Write an equation that can be used to find the value of x.

$2(3x + 1) = 6^2$

9. Solve the equation you wrote in question 8 to find the value of x.

$6x + 2 = 36$
$6x = 34$
$x = 5.\overline{6}$

10. What is the length of $\overline{YX}$?

$3(5.\overline{6}) + 1 = 18$

Geometry
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Lesson 7 Homework

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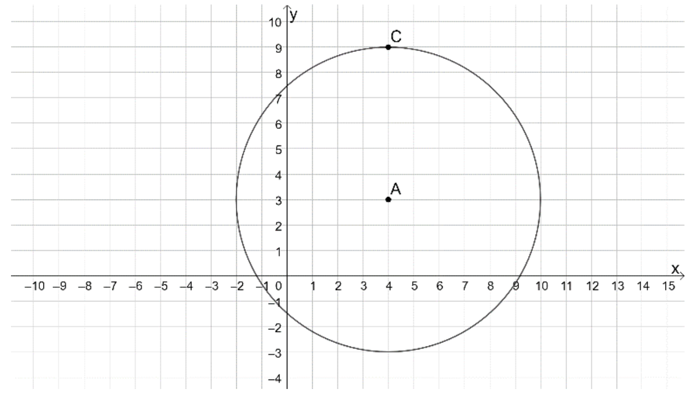
For questions 1 and 2, Circle A is shown. Point C lies on the circle.

1. Give the key features of the circle.



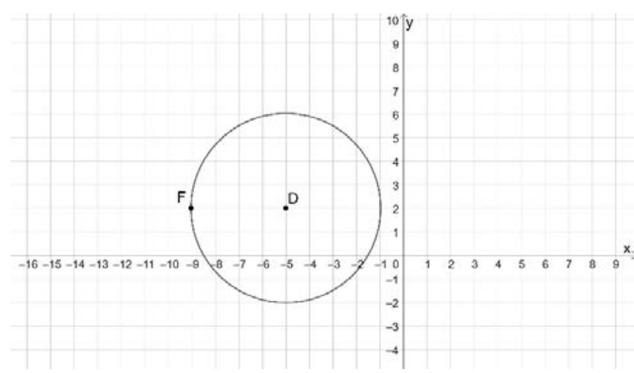
Center	Diameter
$(4, 3)$	12

2. Write the equation that represents Circle A.

$(x-4)^2+(y-3)^2=36$

For questions 3 and 4, Circle D is shown. Point F lies on the circle.

3. Give the key features of the circle.

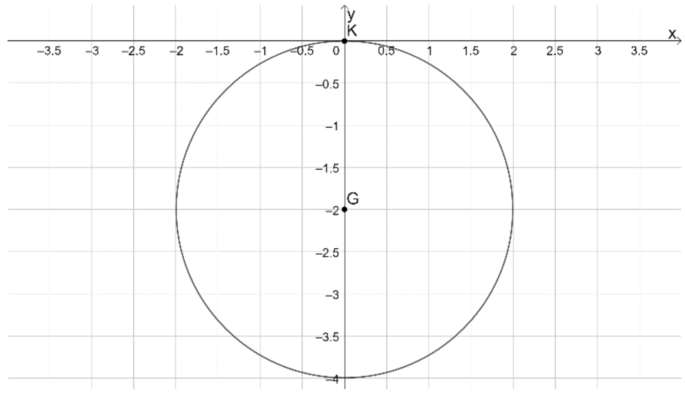


Center	Diameter
$(-5, 2)$	8

4. Write the equation that represents Circle D.

$(x+5)^2+(y-2)^2=16$

For questions 5 and 6, Circle G is shown.
Point K lies on the circle.



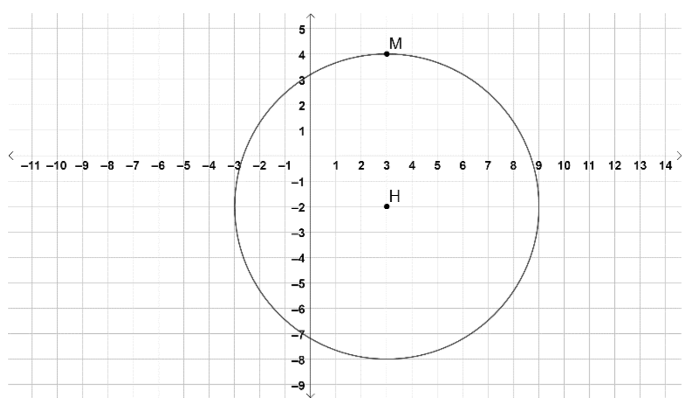
5. Give the key features of the circle.

Center	Diameter
$(0, -2)$	4

6. Write the equation that represents Circle G.

$x^2 + (y + 2)^2 = 4$

For questions 7 and 8, Circle H is shown.
Point M lies on the circle.



7. Give the key features of the circle.

Center	Diameter
$(3, -2)$	12

8. Write the equation that represents Circle H.

$(x - 3)^2 + (y + 2)^2 = 36$

Geometry
Unit 12
Lesson 8 Homework

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Date______________________
Period______________________

For questions 1 and 2, consider the equation of the circle represented by $(x+3)^2 + (y+4)^2 = 9^2$.

1. Rewrite the equation in general form.

$$x^2 + 6x + 9 + y^2 + 8y + 16 = 81$$
$$x^2 + y^2 + 6x + 8y - 56 = 0$$

2. Give the key features of the circle.

<table>
<tr><th>Center</th><th>Diameter</th><th>Radius</th></tr>
<tr><td>$(-3,-4)$</td><td>18</td><td rowspan="3">9</td></tr>
<tr><th>Domain</th><th>Range</th></tr>
<tr><td>$-12 \le x \le 6$</td><td>$-13 \le y \le 5$</td></tr>
</table>

For questions 3 and 4, consider the equation of the circle represented by $(x-5)^2 + (y+2)^2 = 5^2$.

3. Rewrite the equation in general form.

$$x^2 - 10x + 25 + y^2 + 4y + 4 = 25$$
$$x^2 + y^2 - 10x + 4y + 4 = 0$$

4. Give the key features of the circle.

<table>
<tr><th>Center</th><th>Diameter</th><th>Radius</th></tr>
<tr><td>$(5,-2)$</td><td>10</td><td rowspan="3">5</td></tr>
<tr><th>Domain</th><th>Range</th></tr>
<tr><td>$0 \le x \le 10$</td><td>$-7 \le y \le 3$</td></tr>
</table>

For questions 5 and 6, consider the equation of the circle represented by
$x^2 + y^2 - 2x - 6y = -6$.

5. Rewrite the equation in center-radius form.

$$x^2 - 2x + y^2 - 6y = -6$$
$$x^2 - 2x + 1 + y^2 - 6y + 9 = -6 + 1 + 9$$
$$(x-1)^2 + (y-3)^2 = 4$$

6. Give the key features of the circle.

Center	Diameter	Radius
(1,3)	4	2
Domain	**Range**	
$-1 \leq x \leq 3$	$1 \leq y \leq 5$	

For questions 7 and 8, consider the equation of the circle represented by
$x^2 + y^2 - 20y = 21$.

7. Rewrite the equation in center-radius form.

$$x^2 + y^2 - 20y = 21$$
$$x^2 + y^2 - 20y + 100 = 21 + 100$$
$$x^2 + (y-10)^2 = 121$$

8. Give the key features of the circle.

Center	Diameter	Radius
(0,10)	22	11
Domain	**Range**	
$-11 \leq x \leq 11$	$-1 \leq y \leq 21$	

For questions 9 and 10, consider the equation of the circle represented by
$x^2 + y^2 - 10x = -24$.

9. Rewrite the equation in center-radius form.

$$x^2 + y^2 - 10x = -24$$
$$x^2 - 10x + y^2 = -24$$
$$x^2 - 10x + 25 + y^2 = -24 + 25$$
$$(x-5)^2 + y^2 = 1$$

10. Give the key features of the circle.

Center	Diameter	Radius
(5,0	2	1
Domain	**Range**	
$4 \leq x \leq 6$	$-1 \leq y \leq 1$	

Geometry
Unit 12
Lesson 9 Homework

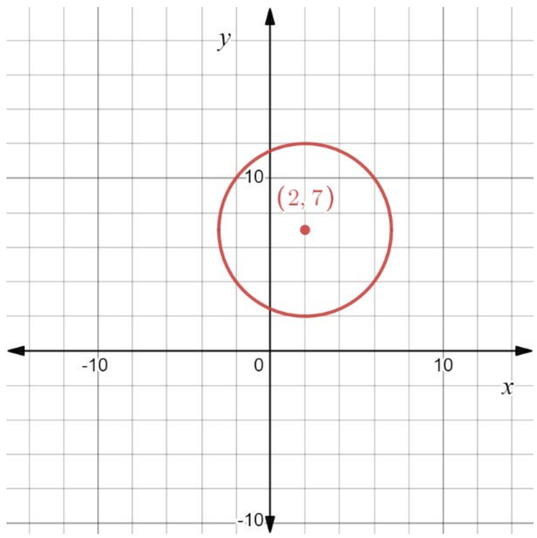
Name Answer Key
Date____________________
Period___________________

For questions 1 and 2, consider that the equation of circle S is $(x-2)^2+(y-7)^2=25$.

1. Give the key features of the circle.

Center	Radius
$(2,7)$	5
Domain	**Range**
$-3 \le x \le 7$	$2 \le y \le 12$

2. Graph circle S.

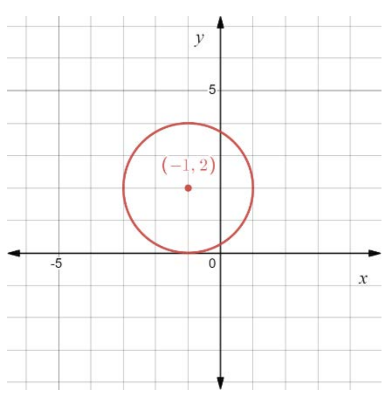


For questions 3 and 4, consider that the equation of circle K is $(x+1)^2+(y-2)^2=4$.

3. Give the key features of the circle.

Center	Radius
$(-1,2)$	2
Domain	**Range**
$-3 \le x \le 1$	$0 \le y \le 4$

4. Graph circle K.



For questions 5 through 7, consider that the equation of circle L is $x^2 + 2x + y^2 + 2y = 7$.

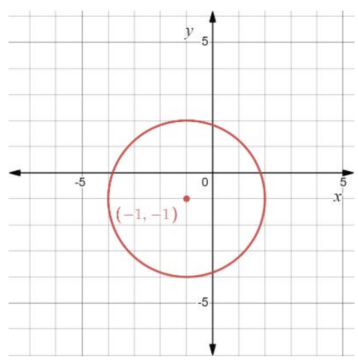
5. Rewrite the equation of the circle in center-radius form.

$$x^2 + 2x + 1 + y^2 + 2y + 1 = 7 + 1 + 1$$
$$(x + 1)^2 + (y + 1)^2 = 9$$

6. Give the key features of the circle.

Center	Radius
$(-1, -1)$	3
Domain	**Range**
$-4 \le x \le 2$	$-4 \le y \le 2$

7. Graph circle .



For questions 8 through 10, consider that the equation of circle Z is

$$x^2 - 6x + y^2 + 4y - 12 = 0.$$

8. Rewrite the equation of the circle in center-radius form.

$x^2 - 6x + 9 + y^2 + 4y + 4 = 12 + 9 + 4$

$(x - 3)^2 + (y + 2)^2 = 25$

9. Give the key features of the circle.

Center	Radius
$(3, -2)$	5
Domain	**Range**
$-2 \le x \le 8$	$-7 \le y \le 3$

10. Graph circle Z.